Semantic Matching between CV and Job Description using NLP and NER

A Comparative Study of TF-IDF, SBERT, Cross-Encoder and NER

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Abstract

Matching a resume to a job description has become a central challenge in modern recruitment pipelines, especially with the widespread use of Applicant Tracking Systems (ATS). This project develops a complete NLP workflow capable of computing semantic similarity between a CV and a job description, while also identifying missing skills using Named Entity Recognition (NER). Several models are compared: TF-IDF, Doc2Vec, SBERT, and a Cross-Encoder. A proportional sampling strategy is applied to balance the job description dataset. Results show that SBERT achieves the highest semantic similarity (68%), while the NER module provides interpretable skill-gap analysis. Beyond numerical performance, this work highlights how contextual embeddings and skill extraction can support fairer and more transparent recruitment processes.

1 Introduction

The volume of job applications received by companies has grown significantly, making manual resume screening both time-consuming and prone to bias. Automated CV–Job Description (JD) matching has therefore become essential for modern recruitment systems. Traditional keyword-based approaches often fail to capture the semantic nuances of professional experience and job requirements.
The objective of this project is to design a robust NLP pipeline capable of:

- computing semantic similarity between a CV and a JD,
- comparing several text representation models,
- extracting and comparing skills using NER,
- providing interpretable feedback to candidates.

The project combines classical vectorization methods with modern transformer-based models, and integrates a skill extraction module to enhance interpretability.

2 Data and Preprocessing

2.1 Datasets

Two Kaggle datasets were used:

- **Job Descriptions**: 1.6M offers with fields such as Job Title, Description, Skills, Responsibilities, and Role.
- **Resumes**: 54k CVs reconstructed from six relational tables (skills, abilities, education, experience, etc.).

2.2 Job Description Preprocessing

The JD dataset was cleaned using:

- lowercasing, punctuation and stopword removal,
- merging of textual fields,
- normalization of role categories.

Since the dataset was too large for efficient processing, we applied **proportional stratified sampling** to obtain a balanced subset of 30,000 job descriptions. This approach preserves the diversity of roles while reducing computational cost.

2.3 Resume Reconstruction

The resume dataset required merging six tables using `person_id`. Each CV was reconstructed by concatenating:

- education entries,
- work experiences,
- skills,
- abilities.

The final dataset contains 54,000 free-text CVs, each representing a realistic professional profile.

3 Methods

3.1 TF-IDF Baseline

TF-IDF provides a lexical baseline. Cosine similarity is computed on 50,000 features. This method is fast but sensitive to vocabulary mismatch and cannot capture paraphrases.

3.2 Doc2Vec

A Doc2Vec model (PV-DM, 300 dimensions) was trained on the full corpus. Although it captures document-level semantics, it is unstable on unseen documents and sensitive to training corpus coverage.

3.3 SBERT

SBERT (`all-MiniLM-L6-v2`) generates dense contextual embeddings. It is robust to paraphrasing and provides efficient semantic similarity computation. It is particularly suited for large-scale retrieval.

3.4 Cross-Encoder

The Cross-Encoder (`stsb-roberta-large`) jointly encodes CV and JD pairs, producing highly accurate similarity scores at the cost of slower inference. It is ideal for re-ranking but not for large-scale search.

3.5 NER Skill Extraction

We used the model `feliponi/hirly-ner-multi` to extract:

- `SKILL`,
- `LANG`.

A two-step comparison was applied:

1. exact matching after Porter stemming,
2. semantic matching using SBERT embeddings (threshold 0.75).

This module provides interpretable insights into which skills are present or missing.

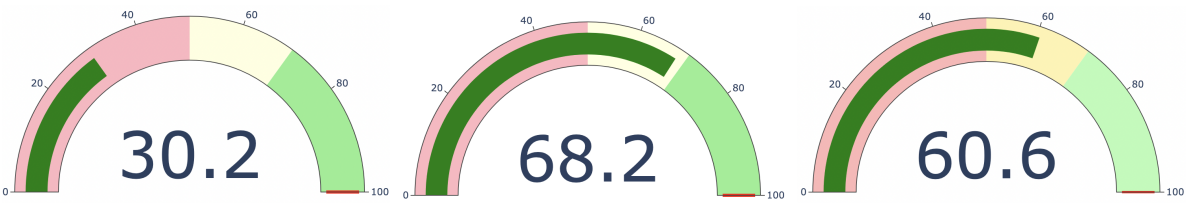
4 Results

4.1 Similarity Scores

Model	Pairwise Score	Top-5 Retrieval
TF-IDF	30.2%	31.04%
Doc2Vec	50%	35%
SBERT	**68%**	**61.56%**
Cross-Encoder	62%	N/A

Table 1: Similarity scores for different models.

4.2 Visual Comparison



(a) TF-IDF (b) SBERT (c) Cross-Encoder

Figure 1: Gauge visualizations of similarity scores.

4.3 NER Skill Comparison

Example:

- CV skills: `ai, algorithms, analytics, api, automation`
- JD skills: `ai, algorithms, consultant, financial services, implementation`

Matched: ai, algorithms
Missing: consultant, financial services, implementation
The NER module provides interpretable feedback for candidates.

5 Discussion

SBERT clearly outperforms other models due to its contextual embeddings.
TF-IDF remains a useful baseline but lacks semantic understanding.
Doc2Vec suffers from instability.
The Cross-Encoder provides strong pairwise accuracy but is unsuitable for large-scale retrieval.
The NER module enhances interpretability by identifying missing skills, although soft-skill extraction remains imperfect.

6 Conclusion

This project demonstrates that combining SBERT with NER yields a powerful and interpretable CV–JD matching system. Future work includes:

- fine-tuning SBERT on HR-specific corpora,
- improving NER for soft skills,
- developing a user-facing interface,
- extending the system to multilingual settings.